

Операционные системы

Анализ файловой структуры UNIX. Команды для работы с файлами и каталогами

Махин Макар Михайлович

2026-09-03

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Выполнить приимеры
- 2 Выполнить дествия по работе с каталогами и файлами
- 3 Выполнить действия с правами доступа
- 4 Получить дополнительные сведения при помощи справки по командам.

2. Процесс выполнения лабораторной работы

```
mmmahin@mmmahin:~$ touch abc1
mmmahin@mmmahin:~$ cp abc1 april
mmmahin@mmmahin:~$ cp abc1 may
mmmahin@mmmahin:~$ mkdir monthly
mmmahin@mmmahin:~$ cp april may monthly
mmmahin@mmmahin:~$ cp monthly/may monthly/june
mmmahin@mmmahin:~$ ls monthly
april  june  may
mmmahin@mmmahin:~$ mkdir monthly.00
mmmahin@mmmahin:~$ cp -r monthly monthly.00
mmmahin@mmmahin:~$ cp -r monthly.00 /tmp
mmmahin@mmmahin:~$
```

Рисунок 1: Выполнение примеров

```
mmmahin@mmmahin:~$  
mmmahin@mmmahin:~$ mv april july  
mmmahin@mmmahin:~$ mv july monthly.00  
mmmahin@mmmahin:~$ ls monthly.00  
july  monthly  
mmmahin@mmmahin:~$ mv monthly.00 monthly.01  
mmmahin@mmmahin:~$ mkdir reports  
mmmahin@mmmahin:~$ mv monthly.01 reports  
mmmahin@mmmahin:~$ mv reports/monthly.01 reports/monthly  
mmmahin@mmmahin:~$
```

Рисунок 2: Выполнение примеров


```
mmmahin@mmmahin:~$  
mmmahin@mmmahin:~$ touch may  
mmmahin@mmmahin:~$ ls -l may  
-rw-r--r--. 1 mmmahin mmmahin 0 Sep  3 13:05 may  
mmmahin@mmmahin:~$ chmod u+x may  
mmmahin@mmmahin:~$ ls -l may  
-rwxr--r--. 1 mmmahin mmmahin 0 Sep  3 13:05 may  
mmmahin@mmmahin:~$ chmod u-x may  
mmmahin@mmmahin:~$ ls -l may  
-rw-r--r--. 1 mmmahin mmmahin 0 Sep  3 13:05 may  
mmmahin@mmmahin:~$ chmod g-r,o-r monthly  
mmmahin@mmmahin:~$ chmod g+w abc1  
mmmahin@mmmahin:~$
```

Рисунок 3: Выполнение примеров

Создание директорий и копирование файлов

```
mmmahin@mmmahin:~$ cp /usr/include/linux/sysinfo.h ~
mmmahin@mmmahin:~$ mv sysinfo.h equipment
mmmahin@mmmahin:~$ mkdir ski.plases
mmmahin@mmmahin:~$ mv equipment ski.plases/
mmmahin@mmmahin:~$ mv ski.plases/equipment ski.plases/equiplist
mmmahin@mmmahin:~$ touch abc1
mmmahin@mmmahin:~$ cp abc1 ski.plases/equiplist2
mmmahin@mmmahin:~$ cd ski.plases/
mmmahin@mmmahin:~/ski.plases$ mkdir equipment
mmmahin@mmmahin:~/ski.plases$ mv equiplist equipment/
mmmahin@mmmahin:~/ski.plases$ mv equiplist2 equipment/
mmmahin@mmmahin:~/ski.plases$ cd
mmmahin@mmmahin:~$ mkdir newdir
mmmahin@mmmahin:~$ mv newdir ski.plases/
mmmahin@mmmahin:~$ mv ski.plases/newdir/ ski.plases/plans
mmmahin@mmmahin:~$
```

Рисунок 4: Работа с каталогами

Работа с командой chmod

```
mmmahin@mmmahin:~$ mkdir australia play
mmmahin@mmmahin:~$ touch my_os feathers
mmmahin@mmmahin:~$ chmod 744 australia/
mmmahin@mmmahin:~$ chmod 711 play/
mmmahin@mmmahin:~$ chmod 544 my_os
mmmahin@mmmahin:~$ chmod 664 feathers
mmmahin@mmmahin:~$ ls -l
total 0
-rw-rw-r--. 1 mmmahin mmmahin  0 Sep  3 13:05 abc1
drwxr--r--. 1 mmmahin mmmahin  0 Sep  3 13:08 australia
drwxr-xr-x. 1 mmmahin mmmahin  0 Sep  3 11:46 Desktop
drwxr-xr-x. 1 mmmahin mmmahin  0 Sep  3 11:46 Documents
drwxr-xr-x. 1 mmmahin mmmahin  0 Sep  3 11:46 Downloads
-rw-rw-r--. 1 mmmahin mmmahin  0 Sep  3 13:08 feathers
drwxr-xr-x. 1 mmmahin mmmahin 74 Sep  3 12:37 git-extended
-rw-r--r--. 1 mmmahin mmmahin  0 Sep  3 13:05 may
drwx--x--x. 1 mmmahin mmmahin 24 Sep  3 13:02 monthly
drwxr-xr-x. 1 mmmahin mmmahin  0 Sep  3 11:46 Music
-r-xr--r--. 1 mmmahin mmmahin  0 Sep  3 13:08 my_os
drwxr-xr-x. 1 mmmahin mmmahin  0 Sep  3 11:46 Pictures
drwx--x--x. 1 mmmahin mmmahin  0 Sep  3 13:08 play
drwxr-xr-x. 1 mmmahin mmmahin  0 Sep  3 11:46 Public
drwxr-xr-x. 1 mmmahin mmmahin 14 Sep  3 13:03 reports
drwxr-xr-x. 1 mmmahin mmmahin 28 Sep  3 13:06 ski.places
drwxr-xr-x. 1 mmmahin mmmahin  0 Sep  3 11:46 Templates
drwxr-xr-x. 1 mmmahin mmmahin  0 Sep  3 11:46 Videos
drwxr-xr-x. 1 mmmahin mmmahin 10 Sep  3 12:06 work
mmmahin@mmmahin:~$
```

```
root:x:0:0:Super User:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/usr/sbin/nologin
dbus:x:81:81:System Message Bus:/usr/sbin/nologin
tss:x:59:59:Account used for TPM access:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/usr/sbin/nologin
polkitd:x:114:114:User for polkitd:/sbin/nologin
systemd-coredump:x:998:998:systemd Core Dumper:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/usr/sbin/nologin
chrony:x:996:996:chrony system user:/var/lib/chrony:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/usr/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/usr/sbin/nologin
unbound:x:993:993:Unbound DNS resolver:/var/lib/unbound:/sbin/nologin
clevis:x:992:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/usr/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/usr/sbin/nologin
gluster:x:991:991:GlusterFS daemons:/run/gluster:/usr/sbin/nologin
qemu:x:107:107:qemu user:/usr/sbin/nologin
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
openvpn:x:990:990:OpenVPN:/etc/openvpn:/usr/sbin/nologin
nm-openvpn:x:989:989:Default user for running openvpn spawned by NetworkManager:/usr/sbin/nologin
/etc/passwd
```

```
mmmahin@mmmahin:~$  
mmmahin@mmmahin:~$ cp feathers file.old  
mmmahin@mmmahin:~$ mv file.old play/  
mmmahin@mmmahin:~$ mkdir fun  
mmmahin@mmmahin:~$ cp -R play/ fun/ I  
mmmahin@mmmahin:~$ mv fun/ play/games  
mmmahin@mmmahin:~$ chmod u-r feathers  
mmmahin@mmmahin:~$ cat feathers  
cat: feathers: Permission denied  
mmmahin@mmmahin:~$ cp feathers feathers2  
cp: cannot open 'feathers' for reading: Permission denied  
mmmahin@mmmahin:~$ chmod u+r feathers  
mmmahin@mmmahin:~$ chmod u-x play/  
mmmahin@mmmahin:~$ cd play/  
bash: cd: play/: Permission denied  
mmmahin@mmmahin:~$ chmod +x play/  
mmmahin@mmmahin:~$
```

Рисунок 7: Работа с файлами и правами доступа

```
MOUNT(8)                                     System Administration                                     MOUNT(8)

NAME
    mount - mount a filesystem

SYNOPSIS
    mount [-h|-V]

    mount [-l] [-t fstype]

    mount -a [-fFnrsvw] [-t fstype] [-O optlist]

    mount [-fnrsvw] [-o options] device|mountpoint

    mount [-fnrsvw] [-t fstype] [-o options] device mountpoint

    mount --bind|--rbind|--move olddir newdir

    mount --make-[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable]
    mountpoint

DESCRIPTION
    All files accessible in a Unix system are arranged in one big tree, the file hierarchy,
    rooted at /. These files can be spread out over several devices. The mount command
    serves to attach the filesystem found on some device to the big file tree. Conversely,
    the umount(8) command will detach it again. The filesystem is used to control how data
    is stored on the device or provided in a virtual way by network or other services.

    The standard form of the mount command is:

        mount -t type device dir

    This tells the kernel to attach the filesystem found on device (which is of type type)
    to the directory dir. (Type 1, press h for help or q to quit)
```

```
Fsck(8)                                     System Administration                                     Fsck(8)

NAME
    fsck - check and repair a Linux filesystem

SYNOPSIS
    fsck [-lsAVRTMNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--]
    [fs-specific-options]

DESCRIPTION
    fsck is used to check and optionally repair one or more Linux filesystems. filesystem
    can be a device name (e.g., /dev/hdc1, /dev/sdb2), a mount point (e.g., /, /usr, /home),
    or a filesystem label or UUID specifier (e.g., UUID=8868abf6-88c5-4a83-98b8-bfc24057f7bd
    or LABEL=root). Normally, the fsck program will try to handle filesystems on different
    physical disk drives in parallel to reduce the total amount of time needed to check all
    of them.

    If no filesystems are specified on the command line, and the -A option is not specified,
    fsck will default to checking filesystems in /etc/fstab serially. This is equivalent to
    the -As options.

    The exit status returned by fsck is the sum of the following conditions:

    0
        No errors

    1
        Filesystem errors corrected

    2
        System should be rebooted

    4
        Operation not supported (fsck will quit)

Manual page fsck(8) Операционные системы q to quit)
```

```

MKFS(8)                                     System Administration      MKFS(8)

NAME
    mkfs - build a Linux filesystem

SYNOPSIS
    mkfs [options] [-t type] [fs-options] device [size]

DESCRIPTION
    This mkfs frontend is deprecated in favour of filesystem specific mkfs.<type> utils.

    mkfs is used to build a Linux filesystem on a device, usually a hard disk partition. The
    device argument is either the device name (e.g., /dev/hda1, /dev/sdb2), or a regular
    file that shall contain the filesystem. The size argument is the number of blocks to be
    used for the filesystem.

    The exit status returned by mkfs is 0 on success and 1 on failure.

    In actuality, mkfs is simply a front-end for the various filesystem builders
    (mkfs.fstype) available under Linux. The filesystem-specific builder is searched for via
    your PATH environment setting only. Please see the filesystem-specific builder manual
    pages for further details.

OPTIONS
    -t, --type type
        Specify the type of filesystem to be built. If not specified, the default filesystem
        type (currently ext2) is used.

    fs-options
        Filesystem-specific options to be passed to the real filesystem builder.

    -V, --verbose
        Produce verbose output, including all filesystem-specific commands that are
        used.

Manual page mkfs(8) (interакционные системы) q to quit)

```



```
KILL(1)                                User Commands                                KILL(1)

NAME
    kill - terminate a process

SYNOPSIS
    kill [-signal|-s signal|-p] [-q value] [-a] [--timeout milliseconds signal] [--]
    pid|name...

    kill -l [number|0xsigmask] | -L

    kill -d pid

DESCRIPTION
    The command kill sends the specified signal to the specified processes or process
    groups.

    If no signal is specified, the TERM signal is sent. The default action for this signal
    is to terminate the process. This signal should be used in preference to the KILL signal
    (number 9), since a process may install a handler for the TERM signal in order to
    perform clean-up steps before terminating in an orderly fashion. If a process does not
    terminate after a TERM signal has been sent, then the KILL signal may be used; be aware
    that the latter signal cannot be caught, and so does not give the target process the
    opportunity to perform any clean-up before terminating.

    Most modern shells have a builtin kill command, with a usage rather similar to that of
    the command described here. The --all, --pid, and --queue options, and the possibility
    to specify processes by command name, are local extensions.

    If signal is 0, then no actual signal is sent, but error checking is still performed.

ARGUMENTS
    The list of processes to be signaled can be a mixture of names and PIDs.
```

3. Выводы по проделанной работе

В ходе данной работы мы ознакомились с файловой системой Linux, её структурой, именами и содержанием каталогов. Научились совершать базовые операции с файлами, управлять правами их доступа для пользователя и групп. Ознакомились с Анализом файловой системы. А также получили базовые навыки по проверке использования диска и обслуживанию файловой системы.